Code:

```
// RED
unsigned long redSum(
  unsigned long N,
  unsigned long *operations)
{
  unsigned long sum = 0;

  *operations = 0;

  for (unsigned long i = 1; i <= N; i++)
  {
    sum = sum + i;
    (*operations)++;
  }
  return sum;
}
// Green
unsigned long greenSum(
  unsigned long N,
  unsigned long *operations)
{
  *operations = 1;

  return (N * (N + 1UL)) / 2UL;
}
// CPU benchmark
void benchmarkCPU(unsigned long N)
{
  unsigned long result;

  unsigned long operations;

  unsigned long redStart;
  unsigned long greenStart;

  unsigned long redTime;
  unsigned long greenTime;

  unsigned long redTotalOperations;
  unsigned long greenTotalOperations;

  volatile unsigned long preventOptimization;
  redStart = micros();

  redTotalOperations = 0;

  for (unsigned long run = 0;
       run < BENCHMARK_RUNS;
       run++)
  {
    result = redSum(N, &operations);

    redTotalOperations += operations;

    preventOptimization = result;
  }

  redTime = micros() - redStart;
  greenStart = micros();

  greenTotalOperations = 0;

  for (unsigned long run = 0;
       run < BENCHMARK_RUNS;
       run++)
  {
    result = greenSum(N, &operations);

    greenTotalOperations += operations;

    preventOptimization = result;
  }

  greenTime = micros() - greenStart
  // ========================================================
  // RED
  // ========================================================

  Serial.println();
  Serial.println("[RED ALGORITHM]");
  Serial.println("--------------------------------");

  Serial.println("Method        : Repeated Addition");
  Serial.println("Complexity    : O(N)");

  Serial.print("Result         : ");
  Serial.println(result);

  Serial.print("Operations/run : ");
  Serial.println(operations);

  Serial.print("Total operations : ");
  Serial.println(redTotalOperations);

  Serial.print("Execution Time : ");
  Serial.print(redTime);
  Serial.println(" us");

  Serial.print("CPU Utilization : ");
  Serial.print(redCPU, 2);
  Serial.println("%");
  Ser
```

Output:

Calculating the Sum of N numbers:

Sketch uses 6496 bytes (20%) of program storage space. Maximum is 32256 bytes.

Global variables use 1424 bytes (69%) of dynamic memory, leaving 624 bytes for local variables. Maximum is 2048 bytes.

Output from Serial Monitor:

```
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    GREEN ALGORITHM LAB

      ARDUINO UNO

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HARDWARE INFORMATION
--------------------------------
Microcontroller : ATmega328P
CPU           : 8-bit AVR
Clock Speed    : 16 MHz
SRAM          : 2 KB
Language       : Arduino C++
TASK:
Calculate SUM(1 … N)

[RED]
Repeated Addition -> O(N)
```

[GREEN]

Mathematical Formula -> O(1)

==

SELECT TEST:

1 -> N = 100

2 -> N = 1,000

3 -> N = 10,000

4 -> N = 50,000

==

Method : Mathematical Formula

Complexity : O(1)

Result : 5050

Operations/run : 1

Total operations : 1000

Execution Time : 1460 us

CPU Utilization : 0.15%

==

RESULT CHECK : PASS

Both algorithms produce the SAME result.

[EFFICIENCY COMPARISON]

RED operations/run : 100

GREEN operations/run : 1

RED CPU time : 140652 us

GREEN CPU time : 1460 us

Operation reduction : 99.00%

[RAM UTILIZATION]

Total SRAM : 2048 bytes

Used SRAM : 1456 bytes

Free SRAM : 592 bytes

RAM Usage : 71.09%

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[GREEN COMPUTING]

Same task

Same result

Fewer operations

Less computational work

Potentially lower energy consumption

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Enter another test:

Enter 1, 2, 3 or 4:

2

==

N = 1000

==

[RED ALGORITHM]

Method : Repeated Addition

Complexity : O(N)

Result : 500500

Operations/run : 1

Total operations : 1000000

Execution Time : 1385536 us

CPU Utilization : 138.55%

[GREEN ALGORITHM]

Method : Mathematical Formula

Complexity : O(1)

Result : 500500

Operations/run : 1

Total operations : 1000000

Execution Time : 1456 us

CPU Utilization : 0.15%

==

RESULT CHECK : PASS

Both algorithms produce the SAME result.

[EFFICIENCY COMPARISON]

RED operations/run : 1000

GREEN operations/run : 1

RED CPU time : 1385536 us

GREEN CPU time : 1456 us

Operation reduction : 99.90%